Adversarial Robustness in Network Intrusion Detection Systems

via Attribution-Guided Feature Selection and Dynamic Multi-Stage Adversarial Training

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Presentation Outline

Overview of the research structure and milestones.

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Problem Statement

Modern Deep NIDS achieve >95% accuracy on normal traffic, but collapse to ~30% under tiny, undetectable packet tweaks.

System Performance

Clean vs. Robust

95% → **~30%**

CLEAN ACC ROBUST ACC

The Adversarial Evasion Mechanism

Attackers slightly alter flow timing and packet sizes.

Consequence: **The AI is tricked into classifying dangerous attacks as normal benign traffic.**

Gradient-Volatile Features

Classifiers rely on weak traffic clues that attackers can easily disguise.

Transient Socket Shortcuts

Models memorize port numbers and IP addresses instead of learning real attack behavior.

Defensive Hardening Trade-Offs

Existing defenses train once on old attacks, making them helpless against adaptive, modern attackers.

Motivation & Objectives

Building fast, accurate, and honest AI defense for live enterprise networks

01

Real-Time Speed

High-speed networks cannot wait: slow detection drops valid packets.

Goal: Keep inference under **0.05 ms** per flow for line-rate speed.

02

Accuracy vs Robustness

Defending models usually causes a 2% to 6% drop in normal accuracy.

Goal: Stop attacks without hurting normal accuracy by removing noisy features.

03

Benchmark Hygiene

Models cheat by memorizing port numbers and IP shortcuts.

Goal: Strictly sanitize shortcuts and enforce zero-leakage partitioning.

Literature Review

Survey of 40 peer-reviewed defenses spanning 2021–2026.

Scope of Survey

40

Peer-Reviewed Publications (2021–2026)

Comprehensive analysis spanning deep learning NIDS, adversarial hardening, and Explainable AI (SHAP).

Macro Empirical Findings Across the Literature

01 Clean Accuracy Trade-Off

Adversarial training on unpruned feature spaces leads to a clean accuracy drop of 2% to 6%, reflecting Tsipras' trade-off.

02 Vulnerability of Tree Ensembles

Tree-based ensembles suffer severe degradation under iterative gradient perturbations (>80% Attack Success Rate).

03 Static Perturbation Staleness

Offline data augmentation exhibits reduced effectiveness against adaptive attacks (16% to 20% evasion reported).

04 Inference Latency Overhead

Runtime XAI verification (85–220 ms/flow) introduces substantial latency overhead relative to real-time line-rate constraints.

05 Shortcut Memorization & Clean-Only Bias

Conventional feature selection methods evaluate features solely on nominal traffic, frequently retaining port numbers that act as fragile back doors under perturbation.

Literature Review (Part 2)

Landmark Peer Benchmarks & Defense Paradigms

1. Static Adversarial

Holla et al., IEEE Access 2025

METHOD

3-layer MLP; clean SHAP top 8; static triad augmentation.

RESULTS

Clean 85% drops to 55% under PGD; defense restores to **88% / 84%** Square.

LIMITATION

Static attack staleness & explicit port retention issues.

2. Dynamic Hardening

Olukola & Rahimi, TIFS 2026

METHOD

4-layer MLP; clean warm up + dynamic PGD with adaptive loss scheduling.

RESULTS

94.8% clean accuracy and **89.4%** robust accuracy (0.08 ms latency).

LIMITATION

Operates across all 78 dimensions (+140% overhead); limited explainability.

3. Explainable AI Gate

Güvenç Paltun et al., 2025

METHOD

DNN with runtime SHAP cosine similarity vs. clean centroids.

RESULTS

95.1% clean accuracy; detects **88.5%** of FGSM/PGD perturbations.

LIMITATION

High runtime latency (85–220 ms per flow); susceptible to scaffolding.

4. Graph Attention IDS

Zeng et al., IEEE IoTJ 2025

METHOD

Dual-module channel-spatial attention over dynamic host communication graphs.

RESULTS

97.4% clean accuracy; **76.2%** robust accuracy under topology evasion.

LIMITATION

Window batch delay (>2.4 ms/flow); vulnerable to dummy-edge spoofing.

Comparative Analysis

Empirical benchmarks across 40 surveyed papers: Every best result has a limitation.

Benchmark Frontier	Published Paper	Best Result	Defense Mechanism	Limitation
Best Clean Accuracy	*Roshan et al. (2023) / Zeng et al. (2025)*	**98.2%** Clean	Standard Deep MLP / Graph Attention	**Zero Defense:** Collapses to <32% under white-box PGD attacks; or incurs 2.4 ms delay.
Best White-Box Robustness	*Olukola & Rahimi (IEEE TIFS 2026)*	**89.4%** PGD_{20}	Multi-Stage Dynamic PGD	**Clean Drop & Overhead:** 2.1% clean penalty; keeps all 78 dimensions (+140% GPU).
Best Static Defense	*Holla et al. (IEEE Access 2025)*	**88.0%** Def	Static Triad Augmentation	**Staleness & Shortcuts:** Offline attacks go stale (16–20% evasion); retained ports.
Best Black-Box Robustness	*Ennaji et al. (IEEE TIFS 2025)*	**88.6%** B-Box	Query Perturbation & Anomaly Tracking	**High Complexity:** Stateful query tracking; vulnerable to adaptive white-box transfer.
Best Certified Defense	*Zhou et al. (IEEE TDSC 2023)*	**Provable L_2**	Randomized Smoothing	**Severe Accuracy Crash:** Clean accuracy dropped from 94.5% → 76.2% (-18.3% penalty).

The Literature Dilemma Across 40 surveyed papers, no existing system achieves high robustness without suffering a severe penalty in clean accuracy, latency, or computational complexity.

Proposed Methodology

Data Sanitization and Screening

01 **Sanitization** — Feature Drop

02 **Partitioning** — Train/Val/Test

03 **MI Screening** — Shannon MI

04 **SMOTE** — Balancing

Data Sanitization

Permanently remove non-generalizable shortcuts: **Src/Dst IP, Ports, Timestamp, Flow ID.**

Result: 69 continuous features remain.

Strict Partitioning

80% Dev Set (5M flows) / 20% Test Set (1.2M flows) locked Day 0.

Dev split: **72% Train / 8% Val.**

Parameters fitted strictly on 72% Train.

Feature Selection

Mutual Information filters 69 down to **top 40 candidates.**

Preserves semantics (unlike PCA).

Accelerates Shapley computation by **>60%.**

Oversampling

Continuous-Only SMOTE applied **strictly within Dtrain** (72% Train).

Balances minority attacks without interpolating discrete flags.

Proposed Methodology: SHAP

Our Approach: Measure feature stability under attack to keep only the 27 most reliable features.

01 Clean-Only Vulnerability

Selecting features only on normal traffic is dangerous: features that look important on clean data are often the easiest for attackers to manipulate.

02 Dual-State Attribution Auditing

Train a baseline model on 40 candidates. Attack it with 20-step PGD, then compare feature importance on normal vs. attacked traffic.

03 Exact Mathematical Stability Score (S_j)

Quantifying individual feature robustness under perturbation:

$$S_j = 1 - \frac{\left|\bar{\phi}_j^{\text{clean}} - \bar{\phi}_j^{\text{adv}}\right|}{\max\left(\bar{\phi}_j^{\text{clean}} + \bar{\phi}_j^{\text{adv}},\ 10^{-10}\right)}$$

04 80% Cumulative Stability-Mass Selection

Rank features descending and select minimal subset m satisfying:

$$\min m \quad \text{s. t.} \quad \sum_{j=1}^{m} S_j \geq 0.80 \cdot S_{\text{total}}$$

27 STABLE FEATURES SELECTED

13 VOLATILE FEATURES PRUNED

Proposed Methodology: Two-Phase Dynamic Training

Curriculum training to mitigate static attack staleness.

Why Two Phases?

Attacking a brand-new model from day one ruins its learning. It must learn normal traffic patterns first before facing adversarial attacks.

PHASE 1: Representation Warmup (Epochs 1–10)

Train exclusively on clean data using standard cross-entropy loss so the model builds solid, accurate classification foundations.

PHASE 2: Dynamic Online Training (Epochs 11–50)

Generate live, fresh PGD attacks for every training batch to defeat evolving attacks without hurting normal accuracy.

Joint Objective: $\mathcal{L}_{\text{total}} = 0.5 \cdot \mathcal{L}_{\text{clean}}(f_\theta(x), y) + 0.5 \cdot \mathcal{L}_{\text{adv}}(f_\theta(x_{\text{adv}}), y)$

Defends against evolving attacks while keeping normal traffic accuracy high.

Dataset & Environment

Experimental configuration and technical architecture parameters

1. Benchmark Dataset

CSE-CIC-IDS 2018 (HuggingFace: dhoogla/csecicids2018): 14 attack classes across 7 profiles; curated Parquet flows replace raw 250 GB dataset.

Total Valid Flow Volume:

6,320,003
flows (post-sanitization).

Partition Breakdown:

- **72% Core Train:** 4.55M flows
- **8% Validation:** 505K flows
- **20% Frozen Test:** 1.26M flows

2. Platform & Stack

GPU Platform:
AMD Radeon RX 7600 (8 GB VRAM, RDNA 3 architecture).

Software Stack:
Python 3.10, PyTorch, ROCm Windows, Adversarial Robustness Toolbox.

Batch Optimization:
Vectorized GPU tensor operations (Batch Size = 2048), mixed-precision (~70 min for 50 epochs).

3. Target Architecture

Topology:
4-Layer Multi-Layer Perceptron (MLP).

Layer Design:
Linear → Batch Normalization → ReLU Activation → Dropout (p = 0.30).

Complexity:
Compact feed-forward pass (~14,000 parameters) ensuring high GPU throughput.

27 → 128 → 64 → 32 → 2

Weekly Research Timeline & Gantt Chart



Current Status (September 10, 2026)

Tasks 1 through 7 are complete (Theoretical formulation, literature audit, data ingestion pipeline, and GPU environment verified). The Pre-Defense Milestone has been successfully achieved.

Immediate Transitions & Next Steps

Immediately transitioning to multi-threat evaluation (Weeks 11–12) and final manuscript submission (Weeks 13–14).

Expected Results & Research Hypotheses

Comprehensive multi-threat evaluation protocol and performance targets.

Clean Accuracy

≥95%

Pruning 13 unstable features mitigates degradation.

Robust Accuracy

88.0% – 90%

PGD-100 (ε=0.05) vs. undefended baselines.

Attack Success Rate

< 12.0%

Under iterative white-box gradient attacks.

Inference Speed

< 0.05 ms

Amortized per flow (preliminary: <0.02 ms).

Multi-Threat Evaluation Suite

First-Order Gradient Attacks
FGSM, PGD20, PGD100, BIM

Parameter-Free & Optimization Attacks
Auto-PGD (APGD-CE), Carlini-Wagner (l∞)

Decision-Based & Black-Box Attacks
Square Attack (query-based), Surrogate Model Transferability

VERIFICATION

Gradient Obfuscation Safeguards
Validate that robustness is genuine and not caused by gradient masking (verifying non-zero input gradients and evaluating black-box transferability).

Conclusion & Immediate Next Steps

Summary of proposed contributions and immediate project execution roadmap

PROPOSED CONTRIBUTIONS

Summary of Key Innovations

Theoretical Contribution

Identifies 27 stable, attack-resistant features using our Adversarial Stability SHAP method.

Architectural Contribution

Introduces two-phase dynamic training to stop attack staleness without sacrificing normal traffic accuracy.

Methodological Hygiene

Permanently removes socket shortcut leakage and enforces strict zero-leakage 72/8/20 partitioning on 6.32M valid flows.

SEPT 11 – OCT 11

Immediate Next Milestones

01 Execute 3-seed experimental runs across all 9 adversarial threat models.

02 Conduct ablation studies: (1) Full 69D vs. 40D MI vs. 27D Stable; (2) Static Triad vs. Dynamic PGD.

03 Complete and submit the final B.Sc. thesis manuscript for October defense.